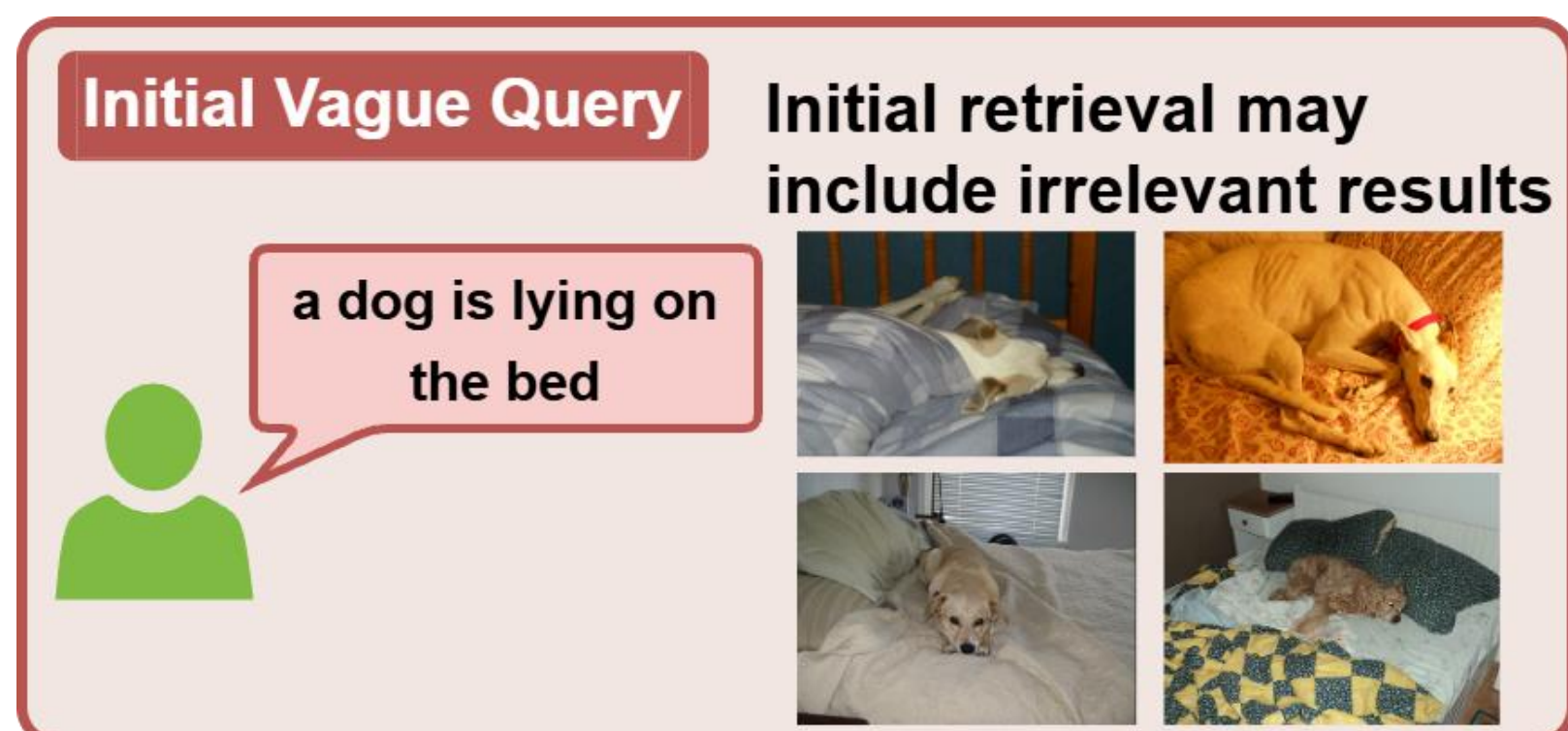


Enable reasoning-guided interactive image retrieval by grounding query refinement in visual facts from retrieved candidates

1 Motivation / Problems

- Retrieval is often treated as **single-step query-image matching**.
- Users sometimes start with **incomplete or ambiguous queries**.
- Users need **guidance** on what **visual details** to add next.

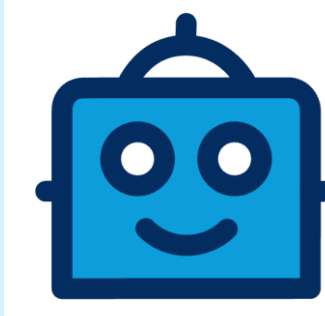


2 Idea

- Use **retrieved candidates** to identify what the current query is missing.
- Represent candidates with **visual facts**: objects, attributes, actions, and relations.
- Select compact evidence using **QVFS**.
- Guide the LLM to **suggest more specific query refinements**.



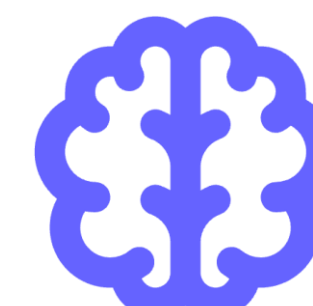
3 Contributions



RAIR-VF framework:
Reasoning-augmented interactive image retrieval



Query-aware Visual Fact Selection (QVFS): Selects compact and informative visual facts.

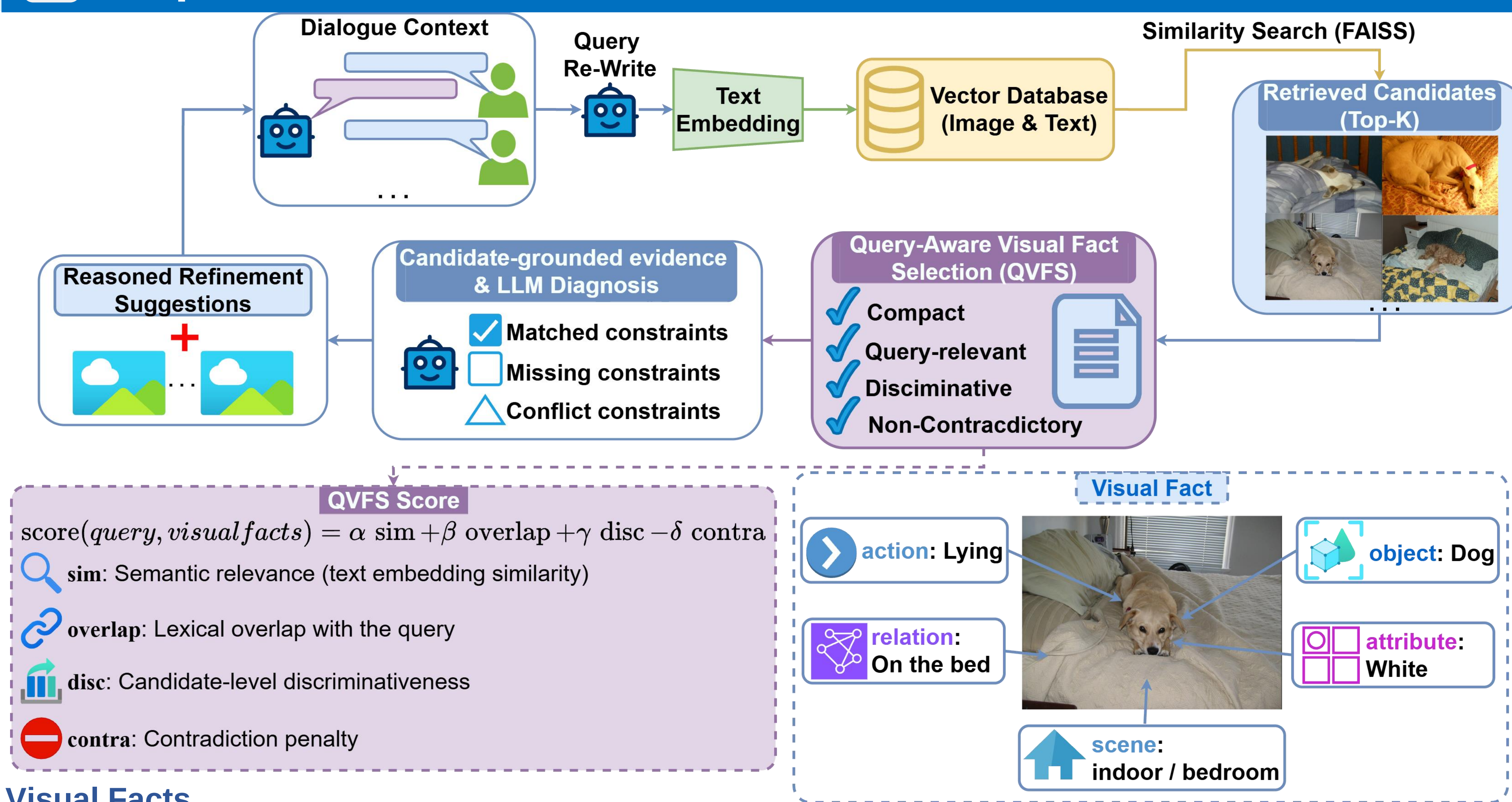


Grounded LLM reasoning:
Diagnoses matched, missing, and conflicting constraints.



Refinement suggestions: Guides users toward more specific queries.

4 Proposed Framework: RAIR-VF



Visual Facts

- Extracted from VisDial annotations and dense-captioning outputs.
- Represent objects, attributes, actions, and relations as short facts.

Query-aware Visual Fact Selection (QVFS)

- Selects compact, query-relevant visual facts.
- Filters out noisy or redundant evidence.

Retrieval Candidates

- Include image, caption, and associated visual facts.
- Serve as grounded context for QVFS and LLM diagnosis.

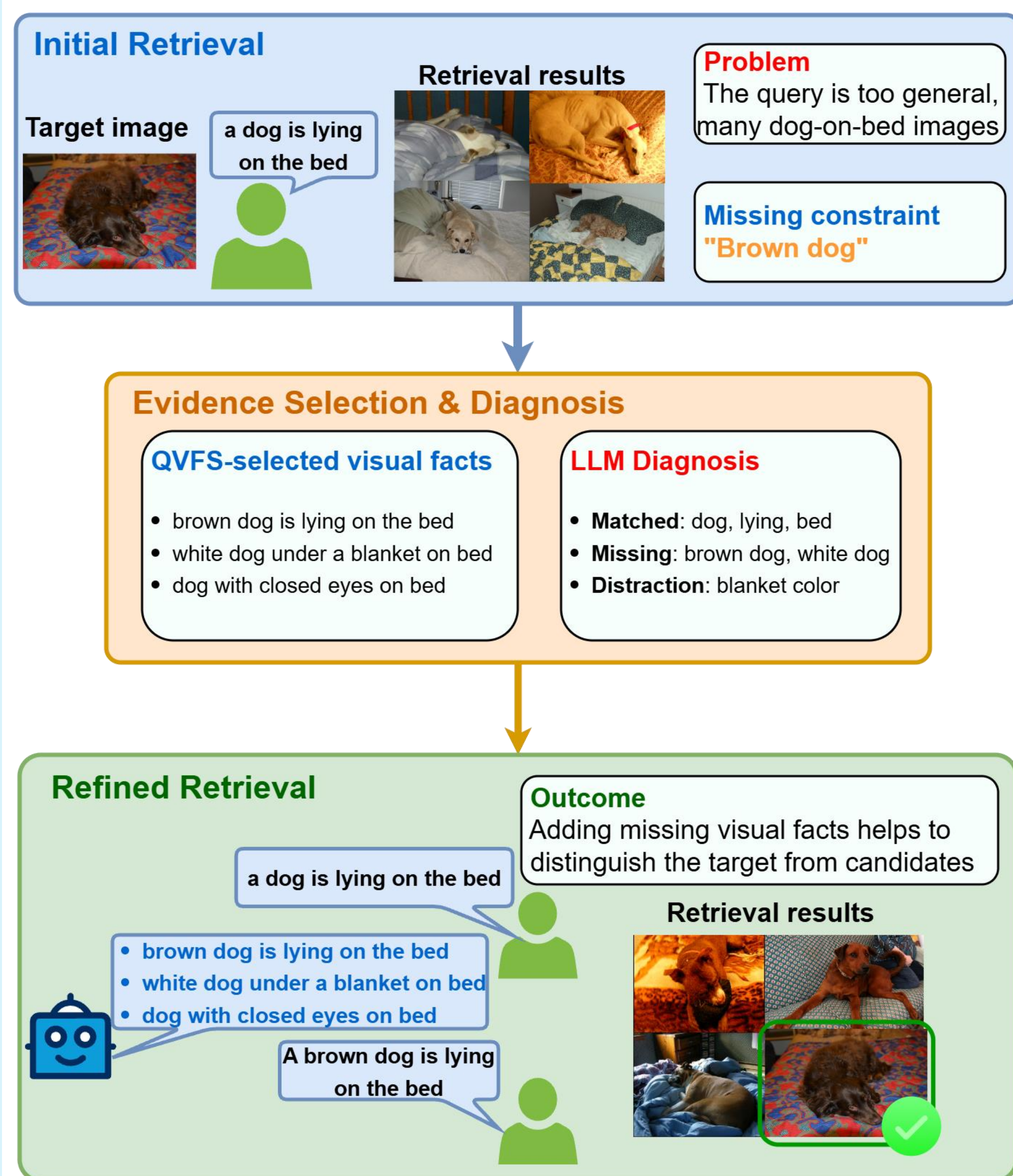
Candidate-grounded Evidence & LLM Diagnosis

- Grounds LLM reasoning in retrieved candidates.
- Identifies matched, missing, and conflicting constraints.

RAIR-VF: Retrieval candidate images, select compact visual facts, diagnose matched / missing / conflicting constraints, and suggest refined queries.

5 Example Interaction

- Initial query is **ambiguous** and retrieves similar **dog-on-bed** images.
- QVFS** extracts useful **visual facts** from **retrieved candidates**.
- LLM diagnosis** detects the missing constraint: **"brown dog"**.
- Refined query** helps rank the target image higher.



6 Experiments & Results

Dataset: Visual Dialog (Visdial) [1]

Metric: Hit@K

Local LLM: Gemma3-12B [2]

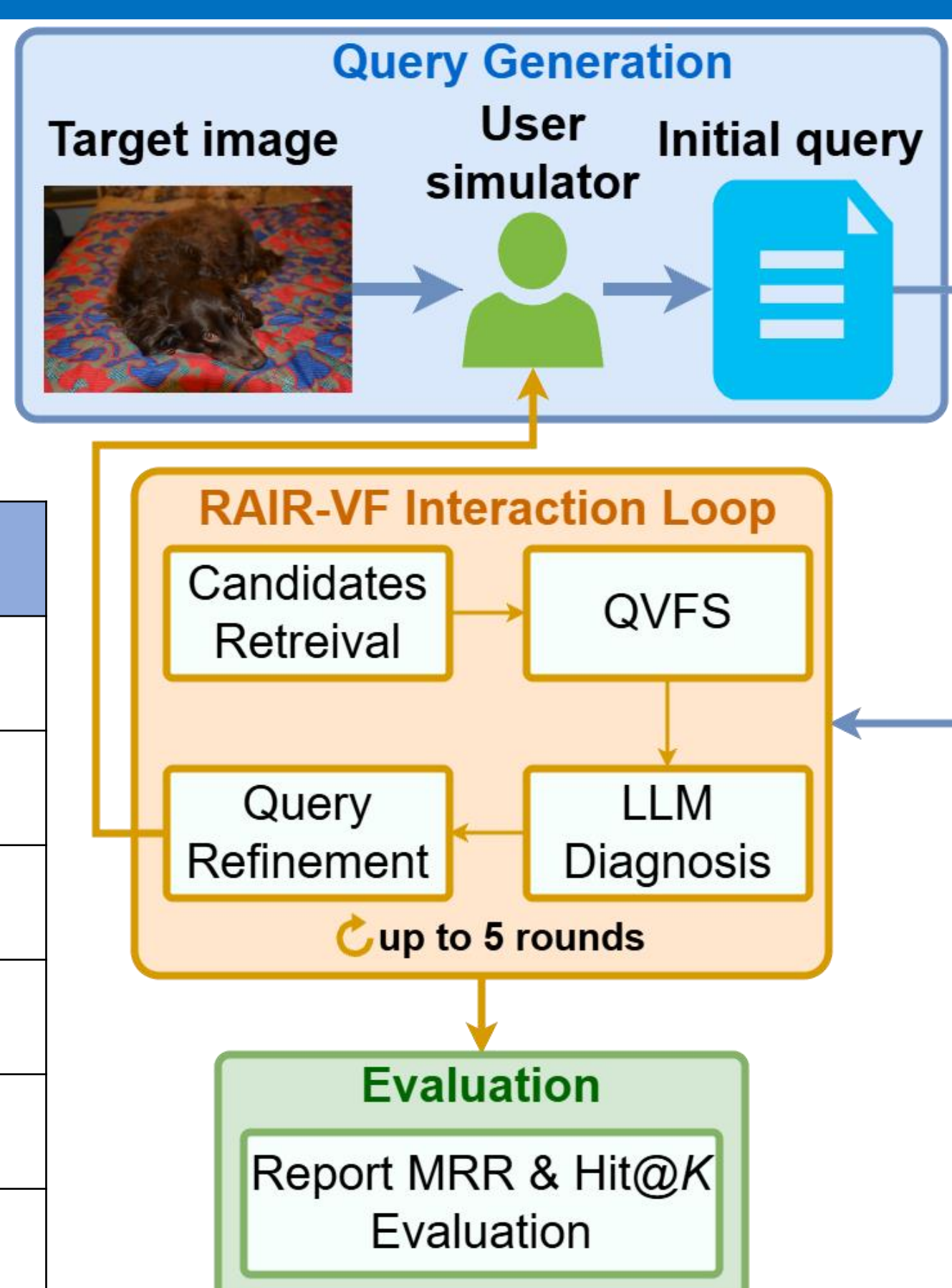
User simulator: Qwen/Qwen2.5-VL-7B-Instruct [3]

Database: PostgreSQL, openai/clip-vit-base-patch32 [4]

Similarity Search: FAISS [5]

Turn	Hit@10	Hit@20	Hit@50
0	0.4444	0.5456	0.6769
1	0.4788	0.5794	0.7125
2	0.4913	0.6394	0.7856
3	0.532	0.6931	0.8212
4	0.5969	0.745	0.8531
5	0.675	0.7756	0.8723

Retrieval improves consistently across interaction turns, with **Hit@10**, **Hit@20**, and **Hit@50** increasing by **+19.06**, **+23.00**, and **+19.54** percentage points, respectively.



7 Conclusion & Future Works

Conclusion

- Candidate-grounded visual facts** support query refinement.
- QVFS** provides compact and relevant evidence.
- LLM diagnosis** helps clarify missing or conflicting constraints.
- Multi-turn interaction** improves best-so-far retrieval performance.

Future Works:

- Extend **RAIR-VF** to larger datasets and more diverse interactive retrieval scenarios.
- Conduct **human-user studies** to evaluate the usefulness and interpretability of refinement suggestions.

[1] A. Das, S. Kottur, K. Gupta, A. Singh, D. Yadav, J. M. F. Moura, D. Parikh, and D. Batra, "Visual Dialog," Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017.

[2] Gemma Team, A. Kamath, J. Ferret, S. Pathak, et al., "Gemma 3 Technical Report," arXiv:2503.19786, 2025.

[3] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge, S. Song, et al., "Qwen2.5-VL Technical Report," arXiv:2502.13923, 2025.

[4] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, et al., "Learning Transferable Visual Models From Natural Language Supervision," Proceedings of the International Conference on Machine Learning (ICML), 2021.

[5] J. Johnson, M. Douze, and H. Jegou, "Billion-scale Similarity Search with GPUs," IEEE Transactions on Big Data, 2019.